Reg. No. | | | | | | | | | | | [14AE662]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

OPERATIONS RESEARCH

Time:3 Hours **Maximum Marks:70**

Instructions:

i. Answer any five full questions selecting at least two full questions from each part.
ii. Missing data may be suitably assumed
iii. Statistical tables will be provided.

PART-A

1. a. The Electro comp Corporation manufactures two electrical products: air conditioners and large fans. The assembly process for each is similar in that both require a certain amount of wiring and drilling. Each air conditioner takes 3 hours of wiring and 2 hours of drilling. Each fan must go through 2 hours of wiring and 1 hour of drilling. During the next production period, 240 hours of wiring time are available and up to 140 hours of drilling time may be used. Each air conditioner sold yields a profit of \$25. Each fan assembled may be sold for a \$15 profit. Formulate and solve this LP production-mix situation, and find the best combination of air conditioners and fans that yields the highest profit. **06**

b. Solve the following Linear Programming Problem by graphical method: **04**

Maximize, $Z = x_1 + x_2$

subject to

$$x_1 + 2x_2 \leq 2000,$$
$$x_1 + x_2 \leq 1500, x_2 \leq 600,$$
$$x_1 \geq 0, x_2 \geq 0$$

c List the phases of operations Research study. **04**

2. a. Solve the following Linear Programming Problem by Simplex method: **07**

Minimize, $Z = x_1 - 3x_2 + 2x_3$

subject to

$$3x_1 - x_2 + 2x_3 \leq 7,$$
$$-2x_1 + 4x_2 \leq 12,$$
$$-4x_1 + 3x_2 + 8x_3 \leq 10$$
$$x_1, x_2, x_3 \geq 0.$$

b. Solve by two phase simplex method. **07**

Minimize $Z = x_1 - 2x_2 - 3x_3$

subject to

$$-2x_1 + x_2 + 2x_3 = 2$$
$$2x_1 + 3x_2 + 4x_3 = 1$$
$$x_1, x_2, x_3 \geq 0$$

3. a. Explain primal-dual relationships in Linear programming problem and hence write the dual of the following Linear programming problem **07**

Maximize $Z = -2x_1 - 2x_2 - 4x_3$

subject to

$$2x_1 + 3x_2 + 5x_3 \geq 2,$$
$$3x_1 + x_2 + 7x_3 \leq 3$$
$$x_1 + 4x_2 + 6x_3 \leq 5$$
$$x_1, x_2, x_3 \geq 0$$

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

b. Solve the following Linear programming problem by Dual Simplex method: **07**

Maximize $Z = -2x_1 - x_2$

subject to

$$3x_1 + x_2 \geq 3$$
$$4x_1 + 3x_2 \geq 6$$
$$x_1 + 2x_2 \geq 3$$
$$x_1, x_2 \geq 0$$

4. a. Obtain an optimal solution to the transportation problem by MODI method as given: **08**

Factory/Warehouse	A	B	C	D	Supply
F	19	30	50	10	7
G	70	30	40	60	9
H	40	8	70	20	18
Demand	5	8	7	14	34

b. A marketing manager has 5 salesman and 5 sales districts. Considering the capabilities of the salesman and the nature of districts, the marketing manager estimates that sales per month (in hundred rupees) for each salesman in each district would be as follows: **06**

	I	II	III	IV	V
A	32	38	40	28	40
B	40	24	28	21	36
C	41	27	33	30	37
D	22	38	41	36	36
E	29	33	40	35	39

Find the assignment of salesman to districts that will result in maximum sale.

PART-B

5. a. Use Revised simplex method to **07**

Maximize $Z = 6x_1 - 2x_2 + 3x_3$

subject to

$$2x_1 - x_2 + 2x_3 \leq 2,$$
$$x_1 + 4x_3 \leq 4,$$
$$x_1 \geq 0, x_2 \geq 0$$

b. Solve the following integer programming problem: **07**

Maximize $Z = 2x_1 + 20x_2 - 10x_3$

subject to

$$2x_1 + 20x_2 + 4x_3 \leq 15,$$
$$6x_1 + 20x_2 + 4x_3 = 20,$$
$$x_1, x_2, x_3 \text{ are non negative integers}$$

6. a. Define: **04**

(i) Saddle point (ii) value of a game (iii) Max-min Principle (v) Pay-off matrix

b. Use Dominance rule to solve the following game: **05**

	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

c. Solve the following game graphically: **05**

	B_1	B_2	B_3
A_1	6	4	3
A_2	2	4	8

7. a. Define critical path. **06**

An assembly operation involves the completion of 11 jobs. The job labels, the time required to complete each one and the necessary immediate predecessors of each job are shown in the table:

Job	Job time(days)	Immediate predecessors
A	13	-
B	08	A
C	10	B
D	9	C
E	11	B
F	10	E
G	8	D,F
H	6	E
I	7	H
J	14	G.I
K	18	J

Draw a network diagram for the project and find the critical path.

b. Assuming that the expected times are normally distributed; find the probability of meeting the scheduled time as given for the network. Scheduled project completion time is 30 days. Also find the date on which the project manager can complete the project with a probability of 0.90. **08**

$[\varphi(0.8944) = 0.8133, \varphi(1.29) = 0.90]$

Activity	Optimistic time (a) (weeks)	Most likely (m) time	Pessimistic (b) time
1-2	2	5	14
1-3	9	12	15
2-4	5	14	17
3-4	2	5	12
4-5	6	6	12
3-5	8	17	20

8. a. Explain essential components of queuing system. **04**

b. Explain Kendall's notation for queuing model. Give description of M/M/1:$(\infty/FIFO)$ model. **05**

c. A drive in bank window has a mean service time of 2 minutes, while the customers arrive at a rate of 20 per hour. Assuming that customers arrive at Poisson distribution and service time follows exponential distribution. **05**

i. What percentage of time will the teller be idle?

ii. After driving up, how long will it take the average customer to wait in line and be served?

iii. What fraction of customers will have to wait in line?